



Scale-invariant batch-adaptive residual learning for person re-identification[☆]

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ABSTRACT

The problem of person re-identification (re-ID) deals with matching two similar persons in probe and gallery sets. The underlying pattern matching task can become more complex as similar persons can appear in different scales in the two sets. In this paper, we address this challenging problem of scale-invariant person re-ID. As a solution, we propose two scale-invariant residual networks with a new loss function for deep metric learning. The first network, termed as Scale Invariant Triplet Network (SI-TriNet), is deeper and is trained from the pre-trained weights. In contrast, the second network, named Scale-Invariant Siamese Resnet-32 (SISR-32), is shallower and uses training from the scratch. Deep metric learning for both the networks are realized through a batch adaptive triplet loss function. Extensive comparisons and ablation studies on the benchmark Market-1501 and CUHK03 datasets clearly demonstrate the effectiveness of the proposed formulation.

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1. Introduction

The problem of person re-identification (re-ID) deals with appearance based matching of pedestrians from two camera views, termed as probe and gallery [1,2]. Re-identification problem becomes difficult to solve as the same person can look very different in these views due to variations in pose, illumination and viewpoint. Furthermore, these persons can also undergo scale variations thereby adding more distortions in their appearance attributes. Variations in scale may occur due to factors like inaccurate localization of a person within a detected bounding box and variations in the physical distance of a person from different cameras in the 3D world. This scale variation greatly increases the complexity of the re-ID task.

Several existing methods have addressed the re-ID problem with two main focus, namely, (a) robust feature descriptor generation [1,3] and (b) better metric learning [4–6]. However, none of these works have explicitly addressed the issue of scale variation. Typically, the existing approaches train a deep Siamese neural network for robust feature extraction. Residual networks (ResNets) [7] have emerged as a popular deep network as they yield comparable performance to other deep models with much less model parameters. Another crucial factor, which plays a key

role in achieving accurate person re-ID, is metric learning. Triplet loss function has become popular for better metric learning [8]. More recently, in [6], the authors use batch hard triplet loss to mine most relevant (hardest) triplets within a batch. However, even such model is found to be susceptible to outliers.

In this paper, we address the problem of scale-invariant person re-ID using scale-invariant residual networks and a new loss function for deep metric learning. Our main contributions are the following: (a) we propose two scale invariant residual architectures and establish that these networks have better gradient activation over conventional ResNets; and (b) we introduce a batch adaptive triplet loss function with better triplet mining capability (over existing batch-hard triplet loss [6]) within a Siamese configuration.

2. Related work

Earlier most of works in re-ID [1,4] are based on hand-crafted system. However, they have failed to yield good results in complex scenarios. Inspired by the excellent performance of the Convolutional Neural Networks (CNNs) in image classification tasks, many recent works in re-ID have explored deep learning as a part of their solution. For example, see the works reported in [2,9–11]. However, accuracy of such models are limited by the unavailability of large training data. Existing deep re-ID models follow a Siamese configuration that compare images from two different sources and return a rank based on a similarity measure. A Siamese network consists of two or more identical (sharing same parameters) sub-networks as pairwise (includ-

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ing two sub-networks) or triplet (including three sub-networks) models. Pairwise Siamese [3,9,10,12] returns a label close to 1 for two similar persons and 0 otherwise. In contrast, Triplet Siamese [2,11,13,14] learns a distance metric that separates similar pairs from dissimilar ones by a appropriate margin.

Now a days, the most efficient way to extract good features is via transfer learning, i.e., fine tuning over pre-trained models (ResNet-50 [7], GoogleNet [15]). Although, such networks outperform most designed models trained from scratch, they often lead to a design lock-in thereby restricting further exploration of new architectures. On the other side, new architectures lack better generalization capability due to limited sample set. Performances of such networks are either improved by data pre-processing/post-processing steps like data augmentation or through complex model designs [3,12,14]. For example, Ristani et al. [13] extracted deep features for multi-target multi-camera tracking and re-identification but employed data augmentation and correlation as pre and post processing steps. Li et al. [10] designed a Harmonious Attention Network (HA-CNN) learning separate loss functions for local and global regions and achieved superior performance with increased model complexity. Modeling of Gated SCNN [3] utilized an additional gate function over the embedding of two persons to achieve person matching while the work of [12] solved an additional sparsity problem over several local and one global channel features to solve joint multi-loss (JLML) classification for person re-ID. Wang et al. [11] added an extra global average pooling layer at the end of each convolution step and fused information from different stages of the network to learn effective features in constrained resource setting. Recently, fine grained visual representation learning [16] is employed in person re-identification tasks for more discriminating representation. In such problem, same persons are distinguished from other similar looking persons based on local attribute information like, gender, handbag, upper color, lower color etc. along with global feature. Some relevant works that are useful in modeling such re-id frameworks are part selection model [17] and object attention model [18]. The fine-grain classification strategy mentioned in these two methods can relatively improve the re-id performance while focusing of important regions to gaze [19] within a detected person for achieving better retrieval among similarly looking persons. Although, these methods have achieved state-of-the-art performance by a relatively simpler model, yet they suffer from a complicated training process. Inclusion of pre-processing and/or post-processing steps, as mentioned above, reduce the actual essence of deep learning in practical usage. We propose two simple scale invariant triplet Siamese models which do not require any form of pre and post-processing. The first one is the modification of the pre-trained ResNet-50 architecture, termed as SI-TriNet. The other one is a much shallower residual network of depth 32, named as SISR-32. These models are much simpler to design in comparison to the above architectures. Additionally, our models can also be integrated into existing as well as new architectures as they do not alter the weights of the network parameters. We have achieved robust recognition performance with these simple scale invariant residual learning networks.

There exist a few works which have achieved metric learning with triplet loss function [2,6,8,11,13,14,20]. Triplet loss is computed over embedding of 3 sets of images, i.e., anchor, positive and negative. While anchor and positive images are from the same person, negative images are from persons different from the anchor. Triplet loss, when applied naively, produces sub optimal results [8] as most of the triplets trivially satisfy the objective and do not contribute to learning. To circumvent this issue, Hermans et al. [6] explored Batch Hard (BH) triplet loss where learning is based only on the hardest positive (farthest distance) sample and the hardest negative (nearest distance) sample with respect

to the anchor within a batch. Although learning with BH triplet loss yields much better results than the traditional triplet loss, it is also susceptible to outliers/wrong annotations. For learning with a better loss, we exponentially decrease the weights for all positives starting from the hardest to the easiest ones. These weights are adjusted in an adaptive manner at each pass of the gradient update process and are thus termed as “Batch Adaptive” triplet loss. This loss will prioritize learning from the hardest positive by also considering the effect of other positives. Among the above-mentioned methods, Zhao et al. [2], Ristani and Tomasi [13], Liu et al. [14] have used original triplet loss function [8] while methods [6,11] have used BH strategy. Li et al. [20] have used a modified form of BH triplet loss which they termed as Squeeze loss for compact cluster formation.

3. Proposed architecture

Feature descriptors (filter kernels) in a CNN are able to detect relevant features irrespective of their spatial locations. However, this behavior (at a given scale) cannot be automatically guaranteed while dealing with more than one scale. Learning feature detectors (filter kernels) that can respond to similar patterns at multiple scales is more likely to improve the recognition task in current deep architectures. Such a neural network can be termed as a scale-invariant convolution neural network (SI-ConvNet) [21].

3.1. Scale-invariant (SI) convolution

In a standard convolution operation, such as the one shown in Fig. 1, output feature maps are generated by convolving an input image/feature x with a set of local feature detectors. These local feature detectors are trainable filters (kernels) parameterized by weights W with spatial extent much smaller than x . The convolved output map f is passed through a nonlinear activation function σ with a bias $b \in \mathbb{R}$. This can be represented as:

$$f = \sigma((W \cdot x) + b) \quad (1)$$

In order to provide scale invariance, input x is spatially transformed to a multi-resolution pyramid instead of learning separate kernels with different spatial extents. This transformed input is convolved with same filter weights and re-scaled back to original dimension. The re-scaling aligns the feature map responses of different resolutions. Finally, a max-pooling operation is performed over this re-scaled feature maps at their corresponding co-located pixel positions to achieve scale-invariant (SI) [21] representation. This obtained feature map is thus robust to scale variations.

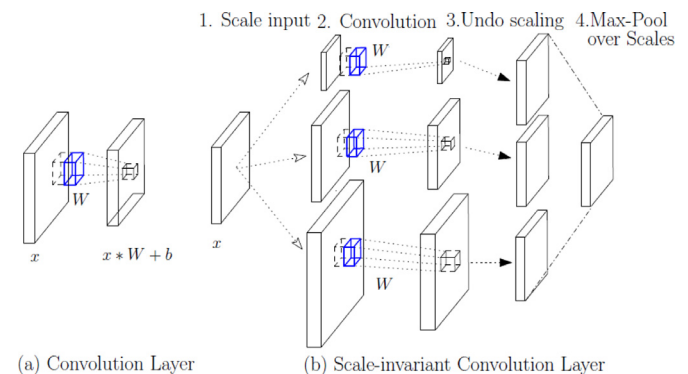


Fig. 1. Comparisons between (a) Convolution layer and (b) Scale-invariant convolution layer [21]. The black box denoted by x is input feature map in both (a) and (b). In (b), several pyramid map of x is constructed (only 3 is shown). Each of these maps is convolved with same kernel filters (W) followed by re-scaling back to original dimension of x . Final convolution map is constructed by max-pooling at corresponding pixel location over all re-scaled maps.

Let, $\mathcal{T}$ be the set of spatial transformations (bilinear interpolation) applied to an input x . Then, for n such transformations, the set $\mathcal{T}$ can be represented as $\{\mathcal{T}_1, \mathcal{T}_2, \dots, \mathcal{T}_n\}$. The scale-invariant feature map f_{SI} [21] can be computed in the following manner:

$$\hat{z}_i = (W \cdot \mathcal{T}_i(x)) + b \quad (2a)$$

$$z_i = \mathcal{T}_i^{-1}(\hat{z}_i) \quad (2b)$$

$$f_{SI} = \sigma \left(\max_{i \in \{1, \dots, n\}} [z_i] \right) \quad (2c)$$

where, $\hat{z}_i \in \mathbb{R}^{h_i \times w_i \times m}$ is the convolution of weights W with the i^{th} spatial transformation of input x (i.e. $\mathcal{T}_i(x)$) each with spatial dimension $h_i \times w_i$ and feature map depth m . Here, $\max(\cdot)$ is the element-wise maximum operation. In Eq. (2b), $z_i \in \mathbb{R}^{h_0 \times w_0 \times m}$ is the output of the inverse transform operation $\mathcal{T}_i^{-1}(\cdot)$ applied over $\hat{z}_i$ to re-scale back to canonical spatial dimension of $h_0 \times w_0$. Without any loss of generality, $\mathcal{T}_1$ is an identity transformation leading to standard convolution. Thus the final output f_{SI} is the maximum of activation at each spatial location. Convolution of a pyramid of input maps with fixed sized (spatial) filter is equivalent to convolving single input with multiple varying sized filters. Scale invariance is achieved based on this equivalence without paying the price of having additional parameters.

3.2. Scale-invariant residual networks

Residual networks (ResNets) [7] have proven to achieve much better results at a much lower network depth (layers) with respect to plain CNN networks. Each ResNet architecture is built from several stacks of “residual units”, as shown in Fig. 2. He et al. [7] have introduced two types of residual units, namely, (a) Baseline residual unit and (b) Bottleneck residual Unit. A residual unit can be expressed in general form as:

$$\mathbf{y}_l = h(\mathbf{x}_l) + \mathcal{F}(\mathbf{x}_l, \mathcal{W}_l) \quad (3a)$$

$$\mathbf{x}_{l+1} = \sigma(\mathbf{y}_l) \quad (3b)$$

where, $\mathbf{x}_l$ and $\mathbf{x}_{l+1}$ are respectively the input and the output of the l^{th} residual unit and $\mathcal{F}(\cdot)$ is the residual function with σ as ReLU [22] activation. In Eq. (3a), $h(\mathbf{x}_l) = \mathbf{x}_l$ would lead to an identity mapping connecting input and output of the residual function $\mathcal{F}(\mathbf{x}_l)$ as shown in Fig. 2. For simplicity, if we ignore the final activation σ in Eq. (3b) and set $h(\mathbf{x}_l)$ to identity mapping then Eqs. (3a) and (3b) can be jointly expressed as

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \mathcal{F}(\mathbf{x}_l, \mathcal{W}_l) \quad (4)$$

Now, during backward propagation, the derivative of $\mathbf{x}_{l+1}$ with respect to $\mathbf{x}_l$ is required, which can be expressed as:

$$\frac{\partial \mathbf{x}_{l+1}}{\partial \mathbf{x}_l} = \mathbf{I} + \frac{\partial \mathcal{F}(\mathbf{x}_l, \mathcal{W}_l)}{\partial \mathbf{x}_l} \quad (5)$$

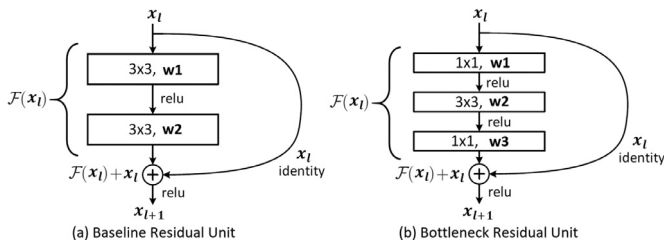


Fig. 2. Two types of residual units introduced by He et al. [7], (a) Baseline Residual Unit and (b) Bottleneck Residual Unit with $\mathcal{F}(\mathbf{x}_l)$ representing two and three stacked convolution layers respectively.

As shown in Fig. 2, the baseline residual block consists of two convolution layers. Suppose the kernel filters in these two layers are parameterized weight matrices $\mathbf{w}_{l,1}$ and $\mathbf{w}_{l,2}$. Then the residual function $\mathcal{F}(\mathbf{x}_l, \mathcal{W}_l)$ can be expressed as:

$$\begin{aligned} \mathcal{F}(\mathbf{x}_l, \mathcal{W}_l) &= \mathcal{W}_l \otimes \mathbf{x}_l \\ &= \mathbf{w}_{l,2} \otimes (\mathbf{w}_{l,1} \otimes \mathbf{x}_l) \\ &= \mathbf{W}_{l,2} \cdot \mathbf{W}_{l,1} \cdot \tilde{\mathbf{x}}_l \end{aligned} \quad (6)$$

where, $\mathbf{W}_{l,1}$ and $\mathbf{W}_{l,2}$ are Toeplitz form of weight matrices $\mathbf{w}_{l,1}$ and $\mathbf{w}_{l,2}$ respectively and $\tilde{\mathbf{x}}_l$ is vectorized form of $\mathbf{x}_l$. The convolution operation can now be represented as matrix-vector multiplication $\mathbf{w}_{l,1} \otimes \mathbf{x}_l = \mathbf{W}_{l,1} \cdot \tilde{\mathbf{x}}_l$. Here, $\mathcal{W}_l = \mathbf{W}_{l,1} \cdot \mathbf{W}_{l,2}$ is equivalence transformation of input $\mathbf{x}_l$ to output $\mathbf{x}_{l+1}$ in the residual unit. Note that we have ignored ReLU activation for simplicity. Thus, combining Eqs. (5) and (6), we can calculate the derivative $\partial \mathbf{x}_{l+1} / \partial \mathbf{x}_l$ using multivariate calculus as:

$$\frac{\partial \mathbf{x}_{l+1}}{\partial \mathbf{x}_l} = \frac{\partial \tilde{\mathbf{x}}_{l+1}}{\partial \tilde{\mathbf{x}}_l} = \mathbf{I} + \mathbf{W}_{l,2} \cdot \mathbf{W}_{l,1} \quad (7)$$

where, $\mathbf{I}$ is the identity matrix. From Eq. (4), we can obtain the output of any deeper unit L from a shallower unit l using:

$$\mathbf{x}_L = \mathbf{x}_l + \sum_{i=l}^{L-1} \mathcal{F}(\mathbf{x}_i, \mathcal{W}_i) \quad (8)$$

It can be easily shown that if the identity connection in Eq. (4) is replaced with a scaling factor λ_l , the net gradient reaching the shallower units will either explode or vanish depending on the magnitude of the scaling factor in back propagation. Let us consider a simple modification in connection $h(\mathbf{x}_l) = \lambda_l \mathbf{x}_l$. Then the equivalence of Eq. (8) is:

$$\begin{aligned} \mathbf{x}_L &= \left(\prod_{i=l}^{L-1} \lambda_i \right) \mathbf{x}_l + \sum_{i=l}^{L-1} \left(\prod_{j=i+1}^{L-1} \lambda_j \right) \mathcal{F}(\mathbf{x}_i, \mathcal{W}_i) \\ &= \left(\prod_{i=l}^{L-1} \lambda_i \right) \mathbf{x}_l + \sum_{i=l}^{L-1} \hat{\mathcal{F}}(\mathbf{x}_i, \mathcal{W}_i) \end{aligned} \quad (9)$$

where, $\hat{\mathcal{F}}(\cdot)$ absorbs scaling factors λ_j into residual functions $\mathcal{F}(\mathbf{x}_i, \mathcal{W}_i)$. Thus, similar to Eq. (7), we can equivalently express gradient $\partial \mathbf{x}_L / \partial \mathbf{x}_l$ as:

$$\frac{\partial \mathbf{x}_L}{\partial \mathbf{x}_l} = \frac{\partial \tilde{\mathbf{x}}_L}{\partial \tilde{\mathbf{x}}_l} = \left(\prod_{i=l}^{L-1} \lambda_i \right) \mathbf{I} + \sum_{i=l}^{L-1} \hat{\mathbf{W}}_{i,2} \cdot \hat{\mathbf{W}}_{i,1} \quad (10)$$

where, $\hat{\mathbf{W}}_{i,1}$, $\hat{\mathbf{W}}_{i,2}$ are Toeplitz form of weight matrices for any intermediate unit i . As evident from the above equation that if $\forall i$, $\lambda_i < 1$, then the first term $(\prod_{i=l}^{L-1} \lambda_i) \mathbf{I} \ll \sum_{i=l}^{L-1} \hat{\mathbf{W}}_{i,2} \cdot \hat{\mathbf{W}}_{i,1}$ will eventually lead to the problem of vanishing gradient. Similarly, if $\forall i$, $\lambda_i > 1$, the same term will lead to the problem of exploding gradient. In our proposed architectures we have replaced each convolution operation in the residual units by SI convolution operation as shown in Fig. 3. Such SI models are robust to scale variation with

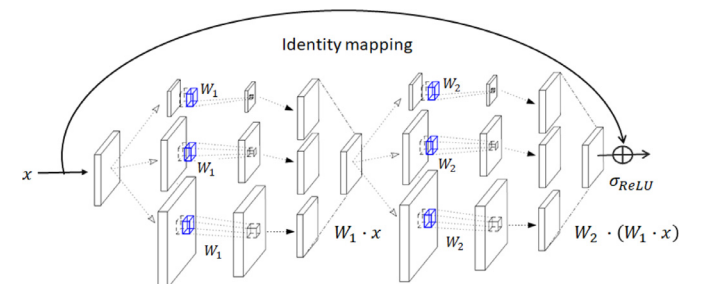


Fig. 3. The proposed SI Residual Unit where normal convolution operation of Baseline Residual Unit (Fig. 3a) is replaced by SI convolution.

improved gradients and do not suffer from vanishing or exploding gradient problems. We now present the following lemma in this connection.

Lemma 1. *The gradients of any deeper unit L with respect to any shallower unit l for scale-invariant (SI) residual network is always greater than or equal to normal residual network.*

Proof. For bilinear interpolation the spatial transform $\mathcal{T}_k(\mathbf{x}_l)$ (for k^{th} scale) can be expressed in linear form $T_k \cdot \bar{\mathbf{x}}_l$. Therefore, Eqs. (2a) and (2b) can be expressed as:

$$\begin{aligned} \mathbf{z}_k^l &= \hat{T}_k \cdot [\mathbf{W}_l \cdot (T_k \cdot \bar{\mathbf{x}}_l) + b_l] \\ &= (\hat{T}_k \cdot \mathbf{W}_l \cdot T_k) \cdot \bar{\mathbf{x}}_l + \hat{T}_k \cdot b_l \\ &= \mathbf{W}_l^k \cdot \bar{\mathbf{x}}_l + \hat{T}_k \cdot b_l \end{aligned} \quad (11)$$

where, T_k and $\hat{T}_k$ are matrices encoding related to k^{th} forward and inverse spatial transform. Here, $\mathbf{W}_l^k$ is equivalent transformation of Toeplitz weight matrix $\mathbf{W}_l$ whose operation is equivalent to Eq. (2b). Therefore, the feature map corresponding to an intermediate SI convolution layer h_{l+1} can be expressed as:

$$\begin{aligned} \tilde{h}_{l+1,SI} &= \widetilde{\max}_{k \in \{1, \dots, n\}} [\mathbf{z}_k^l] \\ &= \widetilde{\max}_{k \in \{1, \dots, n\}} [\mathbf{W}_l^k \cdot \bar{\mathbf{x}}_l + \hat{T}_k \cdot b_l] \\ &= \widetilde{\max}_{k \in \{1, \dots, n\}} [\mathbf{W}_l^k] \cdot \bar{\mathbf{x}}_l + \widetilde{\max}_{k \in \{1, \dots, n\}} [\hat{T}_k] \cdot b_l \end{aligned} \quad (12)$$

Here, we can take $\bar{\mathbf{x}}_l$ and b_l outside $\widetilde{\max}(\cdot)$ as it is a linear operator. Therefore, the gradient of scale-invariant feature map $\partial \mathbf{h}_{l+1}^{SI} / \partial \mathbf{x}_l$ can be represented as:

$$\frac{\partial \mathbf{h}_{l+1,SI}}{\partial \mathbf{x}_l} = \frac{\partial \tilde{h}_{l+1,SI}}{\partial \bar{\mathbf{x}}_l} = \widetilde{\max}_{k \in \{1, \dots, n\}} [\mathbf{W}_l^k] \quad (13)$$

Now, the equivalent form of Eq. (7) for gradient of scale invariant feature map can be represented as:

$$\begin{aligned} \frac{\partial \mathbf{x}_{l+1,SI}}{\partial \mathbf{x}_l} &= \frac{\partial \tilde{\mathbf{x}}_{l+1,SI}}{\partial \bar{\mathbf{x}}_l} \\ &= \mathbf{I} + \left(\widetilde{\max}_{k \in \{1, \dots, n\}} [\mathbf{W}_{l,2}^k] \right) \cdot \left(\widetilde{\max}_{k \in \{1, \dots, n\}} [\mathbf{W}_{l,1}^k] \right) \\ &\geq \mathbf{I} + \mathbf{W}_{l,2} \cdot \mathbf{W}_{l,1} \end{aligned} \quad (14)$$

Please note that both T_k and $\hat{T}_k$ become identity matrices $\mathbf{I}$ for $k = 1$, therefore, $\mathbf{W}_{l,2}^1 = \mathbf{W}_{l,2}$ and $\mathbf{W}_{l,1}^1 = \mathbf{W}_{l,1}$. Therefore, from Eq. (7) and in Eq. (14), we get

$$\frac{\partial \mathbf{x}_{l+1,SI}}{\partial \mathbf{x}_l} \geq \frac{\partial \mathbf{x}_{l+1}}{\partial \mathbf{x}_l} \quad (15)$$

□

Thus, the above formulation shows an improved gradient activation map in SI residual networks.

4. Proposed deep metric learning

Weinberger and Saul [5] proposed metric learning for k-nearest neighbor classification via “Large Margin Nearest neighbor (LMNN)” loss. A problem with LMNN is it cannot properly handle fixed targeted neighbors, FaceNet [8] introduced “triplet loss” [8], which was more suitable for deep metric learning. If g_a , g_p and g_n be an arbitrary anchor, positive and negative (triplet) person set and $D_{a,p}$ and $D_{a,n}$ represents the similarity measure (square euclidean distance) between embedding of pairs (g_a, g_p) and (g_a, g_n) respectively, then a triplet must satisfy the condition $D_{a,p} + \alpha < D_{a,n}$ in learned space. Here α is the margin of separation between positive and negative image pairs. However, this criterion is trivially satisfied for most triplets at initial learning stages which lead

to poor model learning. As a better alternative, Hermans [6] introduced batch hard (BH) triplet loss that is computed based on hardest positive (i.e., $\arg\max_{p \in \{1, \dots, S\}_c, p \neq a} D_{a,p}$) and the hardest negative (i.e., $\arg\min_{n \in \{1, \dots, S\}_c} D_{a,n}$) sample with respect to anchor within a batch. This is expressed as:

$$\mathcal{L}_{BH} = \sum_{c=1}^C \sum_{a=1}^S \left[\alpha + \max_{\substack{p \in \{1, \dots, S\}_c \\ p \neq a}} D_{a,p} - \min_{n \in \{1, \dots, S\}_c} D_{a,n} \right]_+ \quad (16)$$

where, $[\cdot]_+$ is the hinge loss function that stops correcting already satisfied triplets and S is the sample set selected for each C persons (class identities). The hard triplets are mined during training (online) and learning over this triplets are more informative leading to better performance and faster convergence. While BH triplet loss is better in discriminative learning, it is prone to outliers and sometimes fails to detect “normal” associations. We develop an optimal learning strategy to circumvent this problem by designing an adaptive weighting scheme. In this scheme, the positives are weighted based on their hardness (similarity) with respect to the anchor. In our strategy, instead of either equally weighting all positives or entirely weighting the hardest one, we provide weights to the positives based on their distances with the anchor. The hardest (farthest in terms of distance) positive is heavily weighted as compared the easiest (nearest in terms of distance) positive. This is mathematically represented as:

$$\begin{aligned} w_{a,p} &= \frac{\left(\frac{e^{D_{a,p}} - e^{D_{a,p,easy}}}{e^{D_{a,p,hard}} - e^{D_{a,p,easy}}} \right)}{\sum_{\substack{p \in \{1, \dots, S\}_c \\ p \neq a}} \left(\frac{e^{D_{a,p}} - e^{D_{a,p,easy}}}{e^{D_{a,p,hard}} - e^{D_{a,p,easy}}} \right)} \\ &= \frac{\left(\frac{e^{(D_{a,p} - D_{a,p,easy})} - 1}{e^{(D_{a,p,hard} - D_{a,p,easy})} - 1} \right)}{\sum_{\substack{p \in \{1, \dots, S\}_c \\ p \neq a}} \left(\frac{e^{(D_{a,p} - D_{a,p,easy})} - 1}{e^{(D_{a,p,hard} - D_{a,p,easy})} - 1} \right)} \end{aligned} \quad (17)$$

where, $D_{a,p,hard} = \max_{\substack{p \in \{1, \dots, S\}_c \\ p \neq a}} D_{a,p}$ and $D_{a,p,easy} = \min_{\substack{p \in \{1, \dots, S\}_c \\ p \neq a}} D_{a,p}$ are dis-

tances of the farthest and the nearest positives from the anchor respectively. The weights $w_{a,p}$ are also generated online such that every positive sample is dynamically weighted within a batch based on its distance with respect to the anchor. Moreover, the weights are distributed in an exponential fashion from the hardest to the easiest positive with a condition that $\sum_p w_{a,p} = 1$. The main advantage of such a re-weighting technique is once a sample is already pulled closer to the anchor in the embedding space, it gets least weight in the next update. We term this online re-weighting strategy as Batch-adaptive strategy and the corresponding loss as the Batch Adaptive (BA) triplet loss. The proposed Batch Adaptive (BA) loss function is now defined below:

$$\mathcal{L}_{BA} = \sum_{c=1}^C \sum_{a=1}^S \left[\alpha + \frac{1}{S-1} \sum_{\substack{p \in \{1, \dots, S\}_c \\ p \neq a}} (w_{a,p} \cdot D_{a,p}) - \min_{n \in \{1, \dots, S\}_c} D_{a,n} \right]_+ \quad (18)$$

with, C and S are same as described in Eq. (16).

5. Triplet siamese configuration

Deeper neural networks generally perform better than the shallower ones but such networks are difficult to train due to overfitting resulting from unavailability of sufficient data. A better option is to train a network under consideration over the existing pre-trained models. In this work, we propose two SI residual networks, one deeper and one shallower. The deeper SI network is developed from the pre-trained ResNet-50 [7] architecture while the

Table 1
Statistics of Market-1501 and CUHK03 datasets.

Dataset	#Identities (IDs)	#Bounding Boxes (BBs)	#Distractors (BBs)	#Varying Scales (IDs/%)
Market-1501 [23]	1501	32,668	2973 + 500 K	224 (14.92%)
CUHK03 [2]	1476	14,097	0	815 (55.56%)

shallower network is built from stacking several baseline residual units (Fig. 3). Metric learning for both the networks are achieved using the proposed batch adaptive triplet loss function.

5.1. Training from pre-trained weights

We transform ResNet-50 [7] architecture into a scale-invariant network. Triplet Siamese network is adopted to learn the proposed batch adaptive loss function. We name this model Scale invariant Triplet Network (SI-TriNet) architecture. We now state the two important features of this model. Firstly, each sub-network of our SI-TriNet is a scale-invariant modification (standard convolution operations replaced by SI convolutions) of the pre-trained ResNet-50 architecture with initial weights provided by He et al. [7]. Secondly, the global average pooling and softmax (the last two) layers are replaced by a SI convolution layer (to reduce the filter depth down to 512) followed by a global average pooling layer (acting as the final embedding layer). Total number of parameters of this network is 24.64 M.

5.2. Training from scratch

To show the impact of the SI learning over smaller networks, we stack several SI residual units (see Fig. 3) and finally configure it into the triplet Siamese network. This shallower network is termed as Scale-Invariant Siamese Resnet-32 (SISR-32). Within each sub-network each SI residual unit consists of two SI convolution kernels of dimension 3×3 . Five such units are stacked to build a residual block l with fixed kernel depth of n_l for each such block. Three such blocks are cascaded with an increasing filter depth of factor 2 (from $n_1 = 16$ to $n_3 = 64$). Then an average pooling layer of kernel size 8×8 followed by a flattening layer is added for final feature embedding. This is a very light-weight network with network depth 32 and total number of parameters 0.47 M.

6. Experimental results

In this section, we present a brief discussion of the datasets, followed by the evaluation protocol and training details. Then, we show detailed comparisons with several state-of-the-art methods and also include two ablation studies.

6.1. Dataset description

Two widely adopted and publicly available datasets, namely Market-1501 [23] and CUHK03 [2] are chosen for experimentation. Please note that both these datasets include scale variations,



Fig. 4. Example images from person bounding boxes from Market-1501 and CUHK03 datasets showing variations in scales.

as can be seen from Fig. 4. The **Market-1501** [23] dataset contains a total of 32,668 sample images (bounding boxes) of 1501 persons (identities). The images are captured from 6 overlapping views with one having low resolution. The dataset is divided into training (751 identities) and test (750 identities) sets consisting of 12,936 and 19,732 samples respectively. Both single-query (SQ) and multi-query (MQ) evaluations are performed over the dataset. **CUHK03** [2] dataset contains both human-annotated and auto-detected bounding boxes with a total of 1467 identities. The dataset is collected using 6 different cameras with sufficient misalignments and scale variations. We have partitioned the dataset into 867 and 600 identities for training and testing respectively. We have evaluated over both manually annotated (labeled) and detected samples for CUHK03 dataset. Total identities (IDs) with varying scales is 224 (14.92%) in Market-1501 and 815 (55.56%) in CUHK03 dataset. Table 1 shows the detailed statistics of the two datasets.

6.2. Evaluation protocol

The two performance measures [23] widely used for re-ID evaluation are (a) cumulative matching characteristic (CMC) and, (b) mean Average Precision (mAP). The CMC metric measures the percentage of true matches which appear within k ranks over all queries. As re-ID is essentially a ranking problem, rank-1 accuracy is mostly reported in all the recent works. For each query, area under precision recall curve is marked by the average precision (AP). The mean AP, computed over all queries, is denoted by mAP. Ideally, AP is 1 for a query given all true matches are positioned at the top of the rank list. We have performed evaluation on both single and multi-query as described in [23].

6.3. Training details

All the implementation is done using keras API with tensorflow as the backend. For training our SISR-32 model, we feed the original input size, i.e., 128×64 . In contrast, while using pre-trained SI-TriNet, we up-scale the image size to 256×128 to keep the size close to ImageNet classification dataset [7]. Final embedding dimension is kept 512 in both the networks. In scale invariant convolution, the scale factor is of the form 1.26^x [21]. Due to computational cost, we restrict the number of scales to 3 ($1.26^{[-1:1]}$). A batch size of 72 is kept in both the models with $C = 18$ persons and $S = 4$ samples per person at each pass of stochastic gradient updates. We use Adam optimizer with an initial learning rate of 3×10^{-4} and momentum set to the default as given in [6]. We have trained the model for total of $t_1 = 18000$ updates with learning rate decayed by factor $0.001^{(\frac{t-t_0}{t_1-t_0})}$ [6] starting at $t_0 = 12000$. The margin α for triplet loss learning is kept at 0.2 [6].

6.4. Comparison with state-of-the-art methods

Performance of the two proposed models are compared with three baseline methods, KISSME [4], LOMO + Null Space [1] and IDE + ML [25], and ten state-of-the-art methods CAN [14], Gated SCNN [30], JLM [12], TriNet and LuNet [6], Support Neighbor Loss [20], DML [9], HA-CNN [10], MTMCT [13], DaRe [11], MT-net and T-net [2] on Market-1501 dataset. Among these methods, two

Table 2

Performance comparisons on Market-1501 and CUHK03 datasets. Best values are shown in bold.

Methods	Market-1501 Dataset				CUHK03 Dataset			
	Single Query		Multi Query		Detected		Annotated	
	mAP	Rank1	mAP	Rank1	mAP	Rank1	mAP	Rank1
KISSME [4]	12.27	46.72	–	–	–	11.7	–	–
LOMO + Null Space [1]	29.87	55.43	46.03	71.56	–	54.70	–	62.55
CAN [14]	35.9	60.3	47.9	72.1	–	69.2	–	77.6
Gated SCNN [3]	39.55	65.88	48.45	76.04	51.25	61.8	–	–
JLML [12]	64.4	83.9	74.5	89.7	–	80.6	–	83.2
TriNet [6]	69.14	84.92	76.42	90.53	–	–	–	–
LuNet [6]	60.71	81.38	69.07	87.11	–	–	–	–
DML [9]	68.83	87.73	77.14	91.66	–	–	–	–
MTMCT [13]	75.67	89.46	–	–	–	–	–	–
HA-CNN [10]	75.7	91.2	82.8	93.8	38.6	41.7	41.0	44.4
DaRe(R) [11]	69.3	86.4	–	–	51.3	55.1	53.7	58.1
DaRe(De) [11]	69.9	86.0	–	–	50.1	54.3	52.2	56.4
DaRe(R) + RE [11]	74.2	88.5	–	–	58.1	61.6	60.2	64.5
DaRe(De) + RE [11]	76.0	89.0	–	–	59.0	63.3	61.6	66.1
Pose [24]	69.25	87.33	–	–	71.32	78.22	–	–
T-net [2]	50.71	84.26	–	–	–	64.51	–	–
MT-net [2]	62.98	81.59	–	–	–	79.34	–	–
SISR-32 (ours)	51.37	88.19	77.32	93.40	57.13	73.51	61.29	74.91
SI-TriNet (ours)	70.01	96.05	88.08	97	74.39	86.37	76.61	87.32

Table 3

Performance comparison with current state-of-the-art on Market-1501 and CUHK03 dataset with re-ranking. Best values are shown in bold.

Methods	Market-1501 Dataset				CUHK03 Dataset			
	Single Query		Multi Query		Detected		Annotated	
	mAP	Rank1	mAP	Rank1	mAP	Rank1	mAP	Rank1
IDE + ML [25]	63.63	77.11	–	–	–	–	–	–
JLML [12]	72.9	88.8	–	–	–	–	–	–
TriNet [6]	81.07	86.67	87.18	91.75	–	–	–	–
LuNet [6]	75.62	84.59	82.61	89.31	–	–	–	–
SN Loss [20]	86.16	89.90	90.27	93.68	–	–	–	–
DaRe(R) [11]	82.0	88.3	–	–	63.6	62.8	66.7	66.0
DaRe(R) + RE [11]	85.9	90.8	–	–	71.2	69.8	73.7	72.9
DaRe(De) [11]	82.2	88.6	–	–	61.6	60.2	64.1	63.4
DaRe(De) + RE [11]	86.7	90.9	–	–	71.6	70.6	74.7	73.8
SI-TriNet (ours)	82.85	97.38	90.81	98	81.43	94.29	84.33	96.37

baseline methods [1,4] and nine state-of-the-art methods [2,3,6,9–14] reported SQ evaluation, and one baseline method [25] and four [6,11,12,20] recent methods reported MQ evaluation (refer Table 2). Some of these methods [6,11,20,25] have also applied re-ranking strategy to improve their ranking performance (refer Table 3). Out of the two proposed networks, SI-TriNet outperforms all state-of-the-art methods while SISR-32 attains comparable performance with some of the recent methods. SI-TriNet achieves a highest rank-1 accuracy of 96.05% in SQ evaluation when compared to methods like HA-CNN (91.2%), MTMCT (89.46%) and variants of DaRe (highest 89%) and maintains a descent mAP at 70.01%. In MQ evaluation, SI-TriNet achieves highest results in both rank-1 accuracy (97%) and mAP (88.08%) as compared to all other methods. Our SISR-32 model performs decently in rank-1 accuracy (88.19%) and mAP (51.37%) exceeding methods like KISSME (rank-1 46.72% and mAP 12.27%) and LOMO + Null Space (rank-1 55.43% and mAP 29.87%) by large margin. Re-ranking strategy has also boosted the performance of SI-TriNet on both rank-1 accuracy (97.38%) and mAP values (82.85%) for SQ evaluation. For MQ evaluation, SI-TriNet is also highest performer among all other re-ranked methods attaining mAP and rank-1 accuracy of 90.81% and 98% respectively. These results are evident in Table 3.

We have also obtained excellent results in the CUHK03 dataset. Our SI-TriNet model achieves a mAP value of 74.39% and 76.61% exceeding the best performing variants of the popular DaRe [11] by

a margin of 15.39% and 15.01% in detected and annotated sets respectively. The same model has also outclassed the other recent methods like [2,12] by a comfortable margin. Use of re-ranking has further boosted the mAP performance by 7.04% and 7.72% in detected and annotated sets respectively. The SISR-32 model, with only a handful of parameters, also yields competitive performance.

It is observed from the experiments and retrieved results (see Fig. 5 in this connection) that Scale invariant (SI) network generally outperforms other methods if samples with varying scales are more in the dataset in similar settings. We have found CUHK03 and Market-1501 datasets exhibit considerable scale variations. Scale variations between the probe and the gallery set are more prominent in CUHK03 as compared to Market-1501. Please see Table 1 for more details. This is why our model performance in CUHK is better in terms of mean Average Precision (mAP) values (see Tables 2 and 3). In comparison Tables 2 and 3, our methods is slightly lagging (more specifically behind DaRe(De) + RE [11] approach) since they have additionally applied Random Erasing (RE) [11] data augmentation technique to improve their model performance. However, without RE their best performing method, i.e., DaRe(De) achieved a mAP of 69.9% in Market-1501 which is slightly inferior to our proposed model with a mAP value of 70.01%. To show the real impact of variation of scale invariance, we avoided any sort of augmentation in our training process.



Fig. 5. Comparing retrieved samples by Trinet [6] and Our SI-Trinet. Most datasets are retrieved at the beginning of the rank list for all the cases.

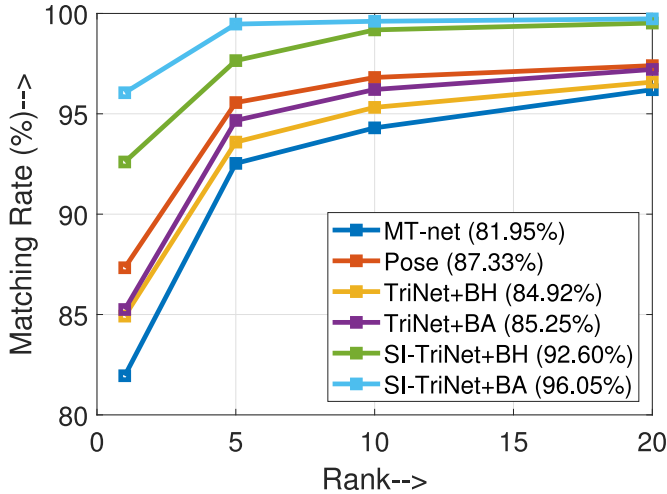


Fig. 6. CMC curves along with Rank-1 recognition rate on Market-1501 dataset.

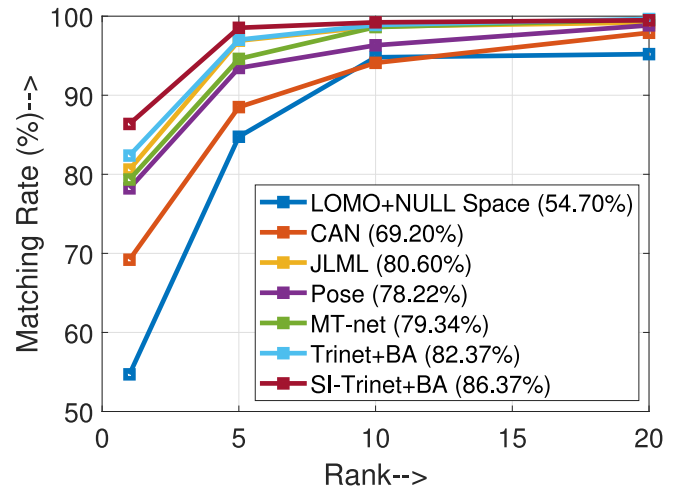


Fig. 7. CMC curves along with Rank-1 recognition rate on CUHK03 dataset.

A detailed analysis along with CMC curves are shown in Figs. 6 and 7 for Market-1501 and CUHK03 dataset respectively. In this figure we have shown how our different methods recognize similar persons within top k ranks, where $k = 1, 5, 10, 15, 20$. From the figure, it can be seen that the overall performance of the proposed SI-TriNet model (rank1: 96.05%) with respect to Normal CNN (rank1: 92.60%) is much better is much better for different Triplet loss functions (i.e., Batch Hard and Batch Adaptive). Fig. 7 also shows similar kind of performance improvement in CUHK03 dataset in which our SI-TriNet model has achieved best performance (rank1: 86.37%). There are two reasons for such improvement. Firstly, most methods have utilized the basic CNN structure in the feature extraction part whereas we have made explicit changes in the architecture (especially the way the Scale invariant (SI) convolution operation has been designed). This leads to layer-wise improvement of the generated feature map resulting in a final embedding which is more robust to scale variations. Secondly, on top of the above SI convolution operation, metric learning is performed with an improved Batch Adaptive Triplet Loss function. Using this function, for each anchor, the loss is com-

puted considering the distance with other positives thereby prioritizing the hardest ones to reduce errors contributed by the outliers. Moreover, the skip connection in SI residual networks leads to a better weight update due to higher gradient activations as compared to usual residual networks (please see Lemma 1 and its proof in Section 3.2 in this connection).

6.5. Ablation studies

In this section we have provided a detailed analysis showing the impact of i) SI convolution over normal convolution and ii) BA triplet loss over BH triplet loss. Table 4 shows the impact of SI convolution over normal convolution both in terms of mAP and rank-1 (with batch adaptive triplet loss in both the cases). The mAP for single query improves from 46.85% to 51.37% and from 71.45% to 77.32% for multi-query in SISR-32 network. Similar trend can also be observed for the SI-TriNet with a jump of 6.44% and 4.32% in mAP for single and multi-query respectively. The above findings clearly indicate the robustness of SI which have significantly boosted the networks' performance. We have also explored the im-

Table 4

SI convolution vs. normal convolution on Market-1501 dataset for single and multi-query.

Methods	Normal(N) Convolution				SI convolution(Ours)			
	Single Query		Multi Query		Single Query		Multi Query	
	mAP	Rank1	mAP	Rank1	mAP	Rank1	mAP	Rank1
TriNet (N/SI)	63.57	85.25	83.71	96.23	70.01	96.05	88.08	97
SR-32 (N/SI)	46.85	76.88	71.45	86.53	51.37	82.19	77.32	93.40

Table 5

Batch-adaptive triplet loss vs. Batch-hard triplet loss on Market-1501 dataset for single and multi-query.

Methods	Batch Hard				Batch Adaptive			
	Single Query		Multi Query		Single Query		Multi Query	
	mAP	Rank1	mAP	Rank1	mAP	Rank1	mAP	Rank1
SI-TriNet	67.58	95.60	86.27	96.1	70.01	96.05	88.08	97
SISR-32	50.79	87.06	75.67	87.87	51.37	88.19	77.32	93.40

fact of BA triplet loss over BH triplet loss (with SI convolution in both the cases). It can be seen from Table 5 that BA triplet loss improves the performance of metric learning over BH. The degree of improvement is more in the deeper SI-TriNet (2.43%) than the shallower SISR-32 (0.58%) in terms of mAP. A possible reason for such difference in improvement is that a shallower network tends to saturate at much lower value as compared to the deeper ones.

7. Conclusion

In this paper, we proposed two scale-invariant residual networks for robust person re-ID tasks. We also introduced a new triplet loss function for better metric learning. Superior performance over the current state-of-the-art approaches on the benchmark Market-1501 and CUHK03 datasets indicate the effectiveness of our formulation. In future, we plan to extend the proposed framework for re-ID problems in open settings.

Declaration of competing interest

We hereby declare that we do not have any conflict of interest for this manuscript.

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